

Covisance: A Real Time Mobile Recolorization Tool for Aiding Color Vision Deficient Users Utilizing D-15 Color Arrangement Test

Gerald R. Tecson
Technological Institute of the Philippines
Quezon City
tecsongeraldr@gmail.com

Gerald T. Cayabyab
Technological Institute of the Philippines
Quezon City
gerald.cayabyab@gmail.com

Fredilyn B. Calanda
Technological Institute of the Philippines
Quezon City
lyn_calanda@yahoo.com

Felizardo C. Reyes Jr.
Technological Institute of the Philippines
Quezon City
felizardo.reyes.jr@tip.edu.ph

ABSTRACT

In this paper, we developed a new mobile application designed to serve as a color vision enhancement tool for individuals with color vision deficiency, or commonly known as color blindness. A person who suffers from Color Vision Deficiency (CVD) faces many complexities in everyday life in even the simplest activities due to having a narrower color spectrum which caused them to perceive colors with similar shades and intensity. Hence, to address this problem, a color vision enhancement method using LMS Daltonization Algorithm was utilized to transform or convert the color of images into a clearer and more recognizable colors to differentiate the colors of two unrelated areas. The application was also designed for a real-time image color processing function that minimizes color discrimination by changing the color for better representation. The application also provided an assessment to identify the user's colorblindness type using D-15 Color Arrangement Test. To validate the accuracy and correctness of information, selected doctors of Optometry and CVD users evaluated the mobile application.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)** → **HCI theory, concepts and models**

Keywords

Color Vision Deficiency; Image Color Processing; Mobile Application; LMS Daltonization Algorithm; Farnsworth D-15 CVD Test

1. INTRODUCTION

Any image or video without colors provide insufficient information to the users. This means vision is the most important human sensor wherein 90% of all information received was through the sense of sight. Around 300 million people across the

world are colour blind. Colour blindness (colour vision deficiency, or CVD) affects approximately 1 in 12 men (8%) and 1 in 200 women [1]. A color vision deficient individual is less sensitive to color contrasts. Thus, they notice fewer details due to narrower color spectrum perceptions. Related researches have been done into the effects of color blindness. This is because until now the general population remains unaware of the difficulties that color blindness can cause on a daily basis. The study focuses on how colorblind people distinguish colors through a mobile application applying enhancing technique (LMS Daltonization algorithm) in real time environment, and utilizing D-15 Disc Arrangement CVD test to identify user's personal type of color vision deficiency.

1.1 Project Objectives

The study aims to design and develop a mobile application that will be used to aid color vision deficient users. Specifically, the study aims to:

- Develop a mobile application that will aid color vision deficient users by applying color enhancement algorithm (LMS Daltonization algorithm) in a real-time environment or an imported image.
- Design mobile application with user-friendly interface by applying concepts of Human Computer Interaction (HCI).
- Implement D-15 CVD Test (Color Arrangement Test) to assess what type of color-blindness the target users have.

2. RELATED LITERATURE

A lot of researches, studies and different approaches or algorithms were significantly proposed and developed. Many previous works have been so devoted and passionate to address CVD compensation. One of these works are from the University of India, Bayuri Sri et al. [2] who developed a mobile application that focuses on designing color blind aid system for Windows Phone7 devices. The mobile app used color blindness Ishihara method and the application of D transform colors using C# programming language library. Test analysis of the system implementation of Ishihara method for co system provides high accuracy result of 100% which is tested by 10 respondents.

Another study that is related to this project is the algorithm developed by Jian Bin et al. [3] where they had developed a fast re-coloring algorithm to improve the accessibility for the color vision impaired. Compared to people with normal color vision, people with color vision impairment have difficulty in distinguishing between certain combinations of colors. To address

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.

CSAI 2017, December 5–7, 2017, Jakarta, Indonesia

© 2017 Association for Computing Machinery.

ACM ISBN 978-1-4503-5392-2/17/12...\$15.00

DOI: <https://doi.org/10.1145/3168390.3168432>

this problem, they had re-mapped the hue components in the HSV color space based on the statistics of local characteristics of the original color image. This approach of enhancing the color contrast through generalized histogram equalization was also used in this study.

The research of Hans et al., [4] entitled “Computerized simulation of color appearance for dichromats” developed a mobile application with a proposed algorithm and transforms a digitized color image to simulate among normal observers the appearance of the image for people who have dichromatic forms of color blindness. The dichromat’s color confusions are deduced from colorimetry, and the residual hues in the transformed image are derived from the reports of unilateral dichromats described in the literature. Like the research of Hans et al, the project represented color stimuli as vectors in a three-dimensional LMS space, and the simulation algorithm was expressed in terms of transformations of this space. The algorithm replaced each stimulus by its projection onto a reduced stimulus surface.

Eduardo et al. [5] proposed a study to find suitable colors based on a clustering algorithm. Asada et al. [6] proposed change in colors based on the colour zone concept to avoid an unnatural image and have developed a tool using smart phone. Chen et al. [7] evaluate preferred images using multiple levels of modified chroma images by colorblind subjects. Wang et al. [8] proposed to change colour to improve accessibility for image content search based on Optimization-Based Colour Rotation and Generalized Histogram Equalization techniques and lastly, Huang et al. [9] proposed how to recolor based on the contrast between two key colors by computing the KL divergence. These studies and researches have explored on the different approaches that can be used for addressing color deficiency.

3. PROJECT DESIGN

Focusing on the main functionalities of this project, below are the project’s features:

Real-Time Recolorization/Color Enhancement using Camera.

This feature requires a good resolution of camera; whereas whenever users’ access their camera they can now enhance the color of the selected scene or object in real time. This technique of recolorization depends on the type of color-blindness of the user. The screen can be divided into split screens: left screen for the daltonize color and right screen for the original color of the scene or object.

Image Recolorization by importing a photo. This feature is an image recoloring or processing method. The user can load a photo from a gallery and transform or convert selected image based on the type of CVD to visualize how color blinded user perceives the image. Converted images can be saved by the app to the generated folder.

D-15 Color Arrangement Test. This feature utilized colorblindness test in a mobile environment that will assess and identify the colorblindness condition of the user by arranging the color palettes provided by the implemented test.

System Architecture. The diagram, illustrated in Figure 1 below, represents the multi-tier mobile architecture of this project composed of three tiers: Presentation Tier, Logic Tier and Data Tier. The presentation tier refers to the applications front-end where the concept of HCI was implemented. The Logic Tier provides the process and procedures to transform all user input

into the desired output. This tier also shows the algorithms and approaches utilized in the study. The third tier or Data Tier includes data persistence mechanism such as file sharing or file storage where the data encapsulates the persistence mechanisms and exposes the resulting data. In the diagram, it presents that all image-processed files will be stored in a generated storage that creates two folders: one for original and one for re-colored image where the algorithm is applied.

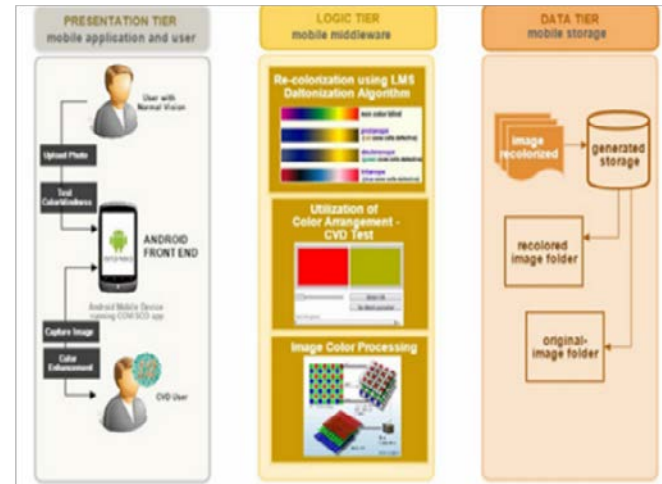


Figure 1. System Architecture.

3.1 Project Development

Methodology. In the development of the project, Figure 2 shows the Rapid Application Development (RAD) methodology, a software development methodology which uses minimal planning in favor of the rapid construction of prototyping, allowing the application to be developed faster by applying rapid changes as required, was applied. This approach is applicable for projects with well-defined objectives with existing data set where decisions can be taken quickly.

3.2 Testing and Operating Procedures

Testing procedure plays a big part on this project development. The results of the tests will scale and validate if the target users met the requirements and satisfaction for this development. The testing standard for this project was based on ISO 9126 Software Quality Characteristics. ISO 9126 is an international standard for the evaluation of software.

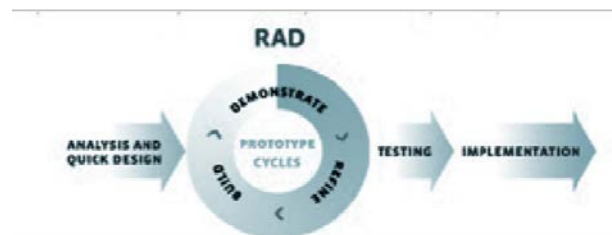


Figure 2. Rapid Application Development Design Model.

Target Users. The application was designed to be used by three types of users: first, users who suffers color vision deficiency; second, users with normal color vision; and third, are the doctor of Optometry or any eye specialists to focus on aiding color blindness.

Development Tools. The project development tool used to make this project was Android Studio, a new android development software designed to provide an integrated environment for building Android applications. We also used Adobe Photoshop CS6 for interface designs and layouts and Microsoft Word 2013 for documentation purposes.

LMS Daltonization Algorithm. This uses an RGB to LMS color space transformation to recolor images from the original color to the converted color. Recoloring, which is the process of changing image colors, is a good example of color transformation or enhancement. Recoloring includes changing colors, intensity, contrast, and brightness of an image. RGB color space is built by adding together three primary colors. The first primary color is near the red portion of the spectrum. The second primary color is near the green of the spectrum whereas the third primary color is near the blue of the spectrum. The basic idea of this algorithm is first to find the LMS values of the RGB (red-green-blue) image using some conversion matrix ("LMS Daltonization Algorithm") [10].

D-15 Color Arrangement Test Algorithm. This is the other type of CVD test which is also known as color arrangement test. There are multiple boxes that represent hues in a certain color and they must be arranged in correct order. A person having color blindness will fail to do this test and make mistakes. Based on this confusion and personal test results, the type of CVD will be calculated. D-15 Color Arrangement Test can differentiate between protan, deutan, and tritan CVD individuals. However, this test can only detect some weak forms of CVD and has the inability to differentiate between Dichromacy and anomalous Trichromacy. This mobile application utilized a more innovative approach to testing user's colorblindness by simply taking this mobile CVD test and this test will first assess the CVD user to verify what type of color-vision deficiency users have. ("Online Farnsworth D-15 Dichotomous Color Blindness Test") [11].

Color Image Processing and Color Enhancement.

Another feature of this mobile application gave the user an accessibility to enhance color vision through an image-recoloring method. The user just simply uploads a photo from a gallery and the mobile application will convert it to an image recolored-photo by applying the same method of enhancement. This project provided a real time re-colorization or color enhancement feature that requires a good resolution of the camera; whereas whenever users access their camera they can now enhance the color of what they perceive that affects their color spectrum in real time. This technique of re-colorization depends on the type of color-blindness of the user. These types are already built in the settings provided for this feature.

4. RESULT AND DISCUSSION

The first phase of the project is gathering of different data about color vision deficiency through in-depth researches in the web and conducting interviews with an expert in Optometry. Different issues, causes, types and tests related to CVD and how they are dealt with were analyzed and studied. Color-blinded people suffered even in the simplest activities and depends on the logical arrangement of colors just like in traffic light and colors of a rainbow. In identifying these issues and problems, the researchers developed a mobile application that will aid users in enhancing their color vision or perceptions. To support the findings, related studies and literature regarding color vision deficiency implemented in mobile applications were analyzed and evaluated to help in the development of Covisance.

For the development phase, identified tools and resources that were essential in the coding part were used to integrate the LMS Daltonization Algorithm in the application. This particular recolorization algorithm is an open-source algorithm that provides mathematical calculations and logical multiplying of different components from an image or any subject. With many types of researches and studies made, adjustments to some details that will help provide a better application were considered in addressing the different needs of the targeted impaired users. RAD software methodology was used and considered in the development of the project.

4.1 Project Description

The mobile application integrated the LMS Daltonization Algorithm to address the problems of users' color deficiency. This process of integration is described in figure 3.

Daltonize. Org [11] discussed that LMS values of the RGB (red-green-blue) image used some conversion matrix. The LMS values correspond to what is received at the level of the retina. The next step is making a conversion delete the information associated with the loss of any of the cone types to get the modified LMS values. Finally, a reverse transformation is done on the LMS values to get the RGB values. RGB presumably represent how that specific color RGB is perceived by a color deficient individual. When this operation is done for all the pixels, the image is converted.

In figure 3 below is a flowchart diagram presenting procedures and conditions on how the study implemented LMS Daltonization Algorithm, from a starting point of receiving input or an image from a camera, to the conversion of RGB coordinates and frames to shifting of wavelengths away from the portion of the spectrum and the displayed image. According to Daltonize.Org [10], there are 8 steps that are followed and performed on the diagram provided above.

LMS Daltonization Algorithm Procedures

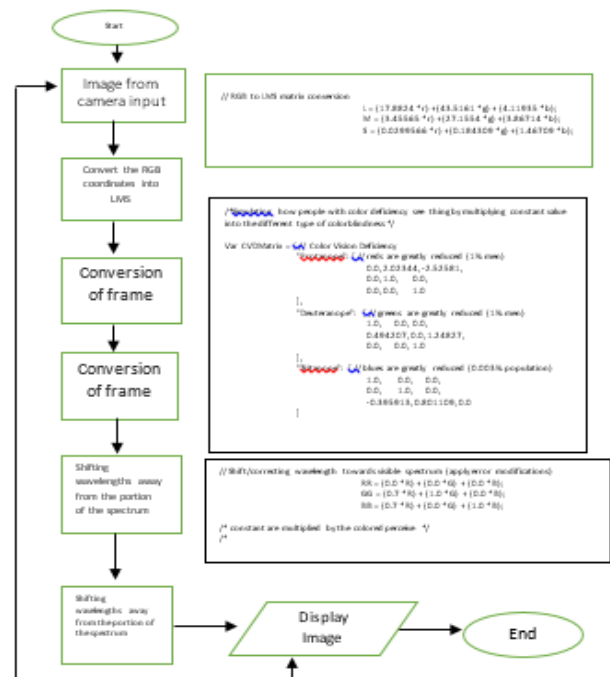


Figure 3. Color Enhancement Procedure using LMS Daltonization Algorithm.

The initial step is basically receiving frame /image from camera input. This is followed by conversion of RGB coordinates into LMS – a color space suitable for calculating color blindness as it's represented by three types of cones of the human eye, named after their sensitivity at wavelength; Long (564-580 nm), Medium (534-545 nm) and Short (420-440 nm)

The variables R, G, B have the color components of inputted image/frame. The numbers 17.8824, 43.5161, 4.11935 are fixed value needed to get Long cone (L) The numbers 3.45565, 27.1554, 3.86714 are fixed value needed to get Medium cone (M) The numbers 0.0299566, 0.184309, 1.46709 are fixed value needed to get Small cone (S).

Hence,

$$\begin{aligned} L &= (17.8824 * r) + (43.5161 * g) + (4.11935 * b); \quad (1) \\ M &= (3.45565 * r) + (27.1554 * g) + (3.86714 * b); \quad (2) \\ S &= (0.0299566 * r) + (0.184309 * g) + (1.46709 * b); \quad (3) \end{aligned}$$

The third step is the actual conversion of frame/ image into a perception of color blindness which is done by reducing the colors along a dichromatic confusion line parallel to the axis of a missing photoreceptor, to a single color.

The following numbers here are the value needed to be multiplied in every CONE with its every component (4), which is programmed as

```
Var CVDMatrix = { // Color Vision Deficiency      (4)
  "Protanopia": [ // reds are greatly reduced (1% men)
    0.0, 2.02344, -2.52581,
    0.0, 1.0, 0.0,
    0.0, 0.0, 1.0 ],
  "Deutanopia": [ // greens are greatly reduced
    (1% men)
    1.0, 0.0, 0.0,
    0.494207, 0.0, 1.24827,
    0.0, 0.0, 1.0 ],
  "Tritanopia": [ // blues are greatly reduced
    (0.003% population)
    1.0, 0.0, 0.0,
    0.0, 1.0, 0.0,
    -0.395913, 0.801109, 0.0 ] };
```

To simulate the color blindness, there is a need to multiply all the values above. If the user clicks the function PROTANOPE, the algorithm will give all the numbers inside of it then multiply it, to get the L,M,S values (5). cvd_means is the first value of the Color vision deficiency that is clicked and so on.

$$\begin{aligned} l &= (cvd_a * L) + (cvd_b * M) + (cvd_c * S); \\ m &= (cvd_d * L) + (cvd_e * M) + (cvd_f * S); \\ s &= (cvd_g * L) + (cvd_h * M) + (cvd_i * S); \end{aligned} \quad (5)$$

The fourth step is LMS to RGB matrix conversion. To convert all the values to RGB (6), there is a need to multiply the values that we get.

$$\begin{aligned} R &= (0.0809444479 * l) + (-0.130504409 * m) + (0.116721066 * s); \\ G &= (-0.0102485335 * l) + (0.0540193266 * m) + (-0.113614708 * s); \\ B &= (-0.000365296938 * l) + (-0.00412161469 * m) + (0.693511405 * s); \end{aligned} \quad (6)$$

Then for step 5, this is for isolating invisible colors to color vision deficiency (calculate error matrix). After getting the RGB components, the difference between the computed RGB to the RGB of the image/frame is done using the formula below:

$$\begin{aligned} R &= r - R; \\ G &= g - G; \\ B &= b - B; \end{aligned}$$

Step six is for shifting colors towards visible spectrum (apply error modifications). RR, GG, BB will have the value that has visible spectrum.

$$\begin{aligned} RR &= (0.0 * R) + (0.0 * G) + (0.0 * B); \\ GG &= (0.7 * R) + (1.0 * G) + (0.0 * B); \\ BB &= (0.7 * R) + (0.0 * G) + (1.0 * B); \end{aligned}$$

The seventh step focused on adding compensation to original values. The computed RR, GG, BB will now be added to the RGB components to make the image/frame shift colors to visible spectrum, then RGB is returned as the sum of the computation. The RGB will be now the new value of the image. The last step is the final display for the Daltonized/enhanced image

$$\begin{aligned} R &= RR + r; \\ G &= GG + g; \\ B &= BB + b; \end{aligned}$$

D-15 Color Agreement Test. Aside from the infamous Ishihara plates, this mobile application preferred D-15 Arrangement CVD Test more, as this test was a subset of the well-known Farnsworth 100 Hue Test. This test is intended for occupational evaluation and classification instead of the in-depth study of color vision defects which contains fifteen numbered hues represented by boxes. Users will arrange these colored boxes and then evaluation of the users' arrangement separates normal color perception from moderate and strong defects in deutan, protan or tritan axis color discrimination.

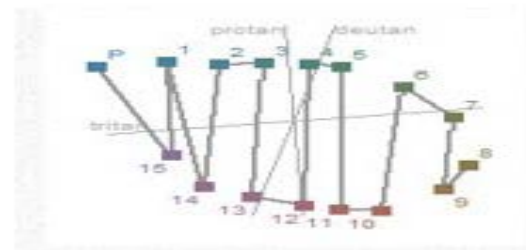


Figure 4. D-15 Color Palette Confusion Lines.

In figure 4 shows how colors are arranged by colorblind people in parallel to one of the three confusion lines: protan, deutan, and tritan. According to Colorblindor [11], it uses D-15 Color Palette Confusion Lines where colorblind people will arrange the colors parallel to one of the three confusion lines. Vingrys and King-Smith developed in 1988 a scoring method based on color difference vectors. This way it is possible to quantify the type of color blindness by your personal confusion angle through the confusion index. The researchers provided a diagram below on how the D-15 CVD Test is performed and calculated. This will show the difference in color arrangement between colorblind people and those without color deficiency. With those confusion lines, the application can tell what type of colorblindness he/she is

suffering from. In d-15 test, only the type of colorblind can be determined but not the severity of his/her colorblindness.

5. SUMMARY & CONCLUSION

The innovation in today's mobility and wireless technology had effectively supported the achievement of the defined objective which is to provide a mobile application tool in aiding color vision impairment. Rapid innovation and mobile technologies and application set a rise in new opportunities and possibilities for the integration mobile services and features covering the aims of this project. The developed mobile application had aided color vision deficient users through mobile test assessments and real-time simulation and recolorization based on the CVD type of assessment.

Color Image Enhancement was achieved through utilization of Daltonization Techniques and Algorithm in a real-time environment. The initial color of the original image was simulated into a perspective of a color vision deficient user through shifting/converting colors. And the result of utilizing a D-15 Color Arrangement Test in mobile significantly contributed to have an accessible and effective assessment of CVD type based from what cones the result of user's final arrangement.

The effectiveness of LMS Daltonization algorithm played a big role in accomplishing the objective functionalities and features of the developed mobile application including the mathematical formulas and computations to process and convert the desired image successfully. The results generated by the mobile application were validated by license optometrist.

6. RECOMMENDATION

The future researches should continue in giving considerations to users whether they are color deficient or not. The designs and interfaces should remain user-friendly in a way that will not discriminate the deficiency of the impaired users. Also, the use of other appropriate algorithms should be tested to enhance the efficiency of assessment and effectiveness of the application.

7. REFERENCES

- [1] <http://www.efds.co.uk/news/3227-recognising-colour-blind-awareness-day-2017-in-sport>
- [2] Bayuri Sri, Baxi, A., & Kotz, D. (2011). Embedded Colorblind Aid System for Windows Phone 7 Devices. Colorblindness Proceedings, the 1998 IEEE International Joint Conference on, 2011.
- [3] Jin Bin, Kahn, Y. Kaski, S. (2009). Recoloring Algorithm to Improve Color Vision Impairment ccessibility. *International journal for eye diseases and impairments*.
- [4] Thus, Hans. Chatii, H., Shahriyar, H., Ruthkauskiene H. (2007). Computerized Simulation of Color Appearance for Dichromats
- [5] Eduardo Macharia. (2011) Suiting Colors using Cluster Algorithm. *Eyes Deficiency in Context*, 10.
- [6] Asada Begona, Ally, M. Gutierrez, S. (2011). Color Changes based Color Zone, Conceptualizations. *Color change Learning. Bulletin of the IEEE Technical Committee on Medical Technology*, 15, 4
- [7] Chen Gros, Dawoud M, K. M. (2011). Modified Chromatic Changes by ColorBlind Subject Preferences. *Colorblindness and Chroma Preference*, 12.
- [8] Wang, Michalson, P. I. (2011) Improvised Access for Image Content Base on Optimization Based Color Rotation and Generalized Histogram Equalization. *GH Equalization and Optimal Color Rotation*
- [9] Huang, Gullmette M., F. I. (2010) Recoloring of Contrast by KL Divergence Computation.
- [10] Daltonize.Org (2010, March 10). LMS Daltonization Algorithm. (Apr. 2015), DOI= <http://www.daltonize.org/2010/05/lms-daltonizationalgorithm.html>
- [11] Colorblindor. (2009, March 10). Online Farnsworth D-15 Dichotomous Color Blindness Test. (Nov. 2015), DOI= <http://www.color-blindness.com/2009/03/10/online-farnsworth-d-15-dichotomous-colorblindness-test/>